



Modelling Financial Markets

Seminar 5: Conditional Volatility — GARCH Models



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator

Acronyms used in this material (2/2)

- **US: United States**
- **UTC: Coordinated Universal Time (Temps Universel Coordonné)**

Outline

Introduction

Part A: Derivations

Part B: Computation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- Run the GARCH variance recursion and its forecasts step by step
- Estimate GARCH, GJR-GARCH and heavy-tailed variants and diagnose the residuals
- Compare models with likelihood ratios, information criteria and out-of-sample losses
- Measure estimation uncertainty with the delta method and a parametric bootstrap, and test on the boundary

- Background: Lecture 5; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- **Part A:** six derivations (A1–A6): multi-step variance, GJR persistence, delta method, moments, robust vs classical QMLE variance, ACF of squares
- **Part B:** eight computational exercises with inference (B1–B8)
- **Part C:** one open question, a seed for the team project (C1), a critique of an AI answer (C2), and a project seed from the lecture case study (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- Report every estimate with its uncertainty: robust standard error, bootstrap interval or p -value
- Check every fitted model: convergence, parameters at a bound, standardised residuals

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (multi-step variance: derivation and numbers), A3 (long-run variance, half-life, delta method)
- Part B: B1 (GARCH-t with diagnostics), B2 (GJR vs GARCH, LR test), B5 (QLIKE, DM, MZ, Giacomini–White, MCS), B7 (parametric bootstrap, IGARCH test under H_0)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A5 (model: A3), A6 (model: A1)
- Part B: B3 (model: B2), B4 (model: B1, B2), B6 (model: B5), B8 (model: B7)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: A3); C3, entropy and Bitcoin risk (model: B1, B2, B5)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter5_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib` and `arch`
- Series: S&P 500, BET, Bitcoin; daily closes up to 18 September 2026
- Returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; equity indices on weekdays, Bitcoin 7 days a week
- Part C joins the BET and S&P 500 prices on common days first, then computes returns

Formulas Used Today

- GARCH(1,1) Bollerslev (1986): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; GJR Glosten, Jagannathan & Runkle (1993): add $\gamma I(\varepsilon_{t-1} < 0) \varepsilon_{t-1}^2$
- Long-run variance $\sigma^2 = \omega / (1 - \alpha - \beta)$; forecast $\mathbb{E}_t \sigma_{t+h}^2 = \sigma^2 + (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \sigma^2)$
- Half-life $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$; $LR = 2(\ell_1 - \ell_0) \sim \chi_m^2$, or $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ for one parameter on the boundary (regular likelihood, identified nuisance parameters)
- Delta method: $\widehat{\text{s.e.}} g(\hat{\theta}) = \sqrt{\nabla g' \hat{V} \nabla g}$; Gaussian QMLE of (ω, α, β) , known mean: $\text{avar} = (\kappa_z - 1)J^{-1}$
- QLIKE loss $r_t^2/h_t + \ln h_t$ Patton (2011); Diebold–Mariano t on the loss differential Diebold & Mariano (1995)
- Sign-bias regression $\hat{z}_t^2$ on S_{t-1}^- , $S_{t-1}^- \hat{\varepsilon}_{t-1}$, $S_{t-1}^+ \hat{\varepsilon}_{t-1}$ Engle & Ng (1993)

A1: Multi-Step Variance Forecast (1/2) [Solved]

Context and data

- ▣ GARCH(1,1): $\varepsilon_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, with z_t i.i.d., mean 0 and variance 1, independent of the past
- ▣ Persistence $p = \alpha + \beta < 1$, long-run variance $\bar{\sigma}^2 = \omega / (1 - p)$; $\mathbb{E}_t$ is the expectation given the information at the close of day t
- ▣ Numbers (returns in %): $\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$, start $\sigma_1^2 = \bar{\sigma}^2$; shocks $\varepsilon_1 = -2.0$, $\varepsilon_2 = 0.5$, $\varepsilon_3 = 3.0$, $\varepsilon_4 = -1.0$

What to do

- ▣ (a) Take conditional expectations of the recursion and derive $\mathbb{E}_t \sigma_{t+h}^2 = \bar{\sigma}^2 + p^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$ for $h \geq 1$.
- ▣ (b) Sum the forecasts and derive the closed form of the H -day variance $V_H = \sum_{h=1}^H \mathbb{E}_t \sigma_{t+h}^2$.
- ▣ (c) Derive the sign of $V_H - H\sigma_{t+1}^2$, the error of the square-root-of-time rule, as a function of $\sigma_{t+1}^2 - \bar{\sigma}^2$.
- ▣ (d) Compute $\sigma_2^2, \dots, \sigma_5^2$ step by step and the forecasts $\mathbb{E}_4 \sigma_{4+h}^2$ for $h = 1, \dots, 5$.

A1: Multi-Step Variance Forecast (2/2) [Solved]

What to do

- (e) Compute V_5 from the sum and from the formula of (b); compare it with $5\sigma_5^2$. (f) Compute the half-life $\ln 0.5 / \ln p$.

What to report and interpret

- Report the two formulas, the four variances, the five forecasts, V_5 with the 5-day volatility $\sqrt{V_5}$ and the half-life.
- Interpretation: after the shock of +3%, does the square-root-of-time rule overstate or understate the 5-day risk, and why?

A1: Multi-Step Variance Forecast: Solution (1/2) [Solved]

- (a) Step 1: shift the recursion, $\sigma_{t+h}^2 = \omega + \alpha \varepsilon_{t+h-1}^2 + \beta \sigma_{t+h-1}^2$, and take $\mathbb{E}_t$.
- Step 2: for $h \geq 2$, $\mathbb{E}_t \varepsilon_{t+h-1}^2 = \mathbb{E}_t[\sigma_{t+h-1}^2 \mathbb{E}(z_{t+h-1}^2)] = \mathbb{E}_t \sigma_{t+h-1}^2$, hence $\mathbb{E}_t \sigma_{t+h}^2 = \omega + p \mathbb{E}_t \sigma_{t+h-1}^2$.
- Step 3: subtract $\bar{\sigma}^2 = \omega + p\bar{\sigma}^2$; the gap to the long run shrinks by p per step, and iterating from the known σ_{t+1}^2 gives the formula.
- (b) Summing, $V_H = H\bar{\sigma}^2 + (\sigma_{t+1}^2 - \bar{\sigma}^2) \sum_{j=0}^{H-1} p^j = H\bar{\sigma}^2 + (\sigma_{t+1}^2 - \bar{\sigma}^2) \frac{1-p^H}{1-p}$.
- (c) Subtracting $H\sigma_{t+1}^2$: $V_H - H\sigma_{t+1}^2 = (\bar{\sigma}^2 - \sigma_{t+1}^2) \left[H - \frac{1-p^H}{1-p} \right]$, and the bracket equals $\sum_{j=0}^{H-1} (1-p^j) > 0$ for $H \geq 2$.
- Interpretation: the rule $H\sigma_{t+1}^2$ understates the H -day variance when today is calmer than the long run and overstates it after turbulent days, because the variance reverts to $\bar{\sigma}^2$.

A1: Multi-Step Variance Forecast: Solution (2/2) [Solved]

- (d) Start: $\sigma_1^2 = 0.05/(1 - 0.95) = 1.0000$; then
 $\sigma_2^2 = \omega + \alpha \varepsilon_1^2 + \beta \sigma_1^2 = 0.05 + 0.10 \times (-2.0)^2 + 0.85 \times 1.0000 = 0.05 + 0.4000 + 0.8500 = 1.3000$.
- In the same way $\sigma_3^2 = 0.05 + 0.0250 + 1.1050 = 1.1800$, $\sigma_4^2 = 0.05 + 0.9000 + 1.0030 = 1.9530$ and
 $\sigma_5^2 = 0.05 + 0.1000 + 1.6600 = 1.8100$.
- Forecasts: the gap $\sigma_5^2 - \bar{\sigma}^2 = 0.8101$ shrinks by 0.95 per day, so $\mathbb{E}_4 \sigma_{4+h}^2 = 1.8100, 1.7695, 1.7311, 1.6945, 1.6598$ for $h = 1, \dots, 5$.
- (e) Formula: $V_5 = 5 \times 1.0000 + 0.8101 \times (1 - 0.95^5)/(1 - 0.95) = 5 + 0.8101 \times 4.524 = 8.665$, the same as the sum;
 5-day volatility $\sqrt{8.665} = 2.944\%$.
- The square-root-of-time rule gives $5\sigma_5^2 = 9.050 > V_5$: after the turbulent days it overstates the 5-day variance, as (c) predicts for $\sigma_5^2 > \bar{\sigma}^2$.
- (f) Half-life = $\ln 0.5 / \ln 0.95 = -0.6931 / (-0.0513) = 13.5$ days: half of any variance shock is gone after about two weeks of trading.

A2: GJR Persistence with Skewed Innovations (1/2) [Proposed]

Solution discussed in class

Context and data

- GJR-GARCH(1,1) [Glosten, Jagannathan & Runkle \(1993\)](#): $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, $I_{t-1} = 1(\varepsilon_{t-1} < 0)$; z_t i.i.d., mean 0, variance 1, not necessarily symmetric
- Data for (c): S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100(\ln P_t - \ln P_{t-1})$; GJR-GARCH(1,1) with skewed-t innovations [Hansen \(1994\)](#): $\hat{\alpha} = 0.00000$, $\hat{\gamma} = 0.21116$, $\hat{\beta} = 0.87965$, $\hat{\eta} = 7.70$, $\hat{\lambda} = -0.150$

What to do

- (a) Take expectations of the recursion and show that the persistence is $\alpha + \gamma m_- + \beta$, with $m_- = \mathbb{E}[z_t^2 1(z_t < 0)]$.
- (b) Show that $m_- = 1/2$ when z_t is symmetric, and explain why m_- differs in general from $P(z_t < 0)$.
- (c) Compute m_- and $P(z_t < 0)$ by numerical integration of the fitted skewed-t density; compute the persistence and the half-life with m_- and with $1/2$.

A2: GJR Persistence with Skewed Innovations (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report m_- , $P(z_t < 0)$, both persistences and both half-lives. Interpretation: what goes wrong if the software formula $\alpha + \gamma/2 + \beta$ is used with skewed innovations?
- Model: A1

A3: Long-Run Volatility, Half-Life and Delta Method (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- GARCH(1,1) with Student-t innovations (Lecture 5): $\hat{\omega} = 0.015714$, $\hat{\alpha} = 0.122573$, $\hat{\beta} = 0.872139$; robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$ from the fit, with $\widehat{\text{s.e.}}(\hat{\alpha} + \hat{\beta}) = 0.0047$; 251.4 trading days per year

What to do

- (a) Derive the long-run variance $\sigma^2 = \omega / (1 - \alpha - \beta)$ by taking unconditional expectations in the recursion.
- (b) Compute the long-run daily and annual volatility and the half-life; compare with the sample volatility 19.2% p.a.
- (c) Compute the gradient of $\sigma^2(\omega, \alpha, \beta)$ and, by the delta method, the standard error of the annual long-run volatility.
- (d) Compute the delta-method standard error of the half-life $h = \ln 0.5 / \ln p$, $p = \alpha + \beta$, and build a 95% interval for h that respects $h > 0$.

A3: Long-Run Volatility, Half-Life and Delta Method (2/2) [Solved]

What to report and interpret

- Report the long-run volatility with its standard error, the half-life with its standard error and the transformed interval.
- Interpretation: why is the delta method useless for the half-life, and what does that mean for the question $p = 1$ (B7)?

A3: Long-Run Volatility, Half-Life and Delta Method: Solution (1/2) [Solved]

- (a) Under covariance stationarity $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2 = \mathbb{E}\varepsilon_{t-1}^2 = \sigma^2$, so $\sigma^2 = \omega + (\alpha + \beta)\sigma^2$ and $\sigma^2 = \omega/(1 - \alpha - \beta)$.
- (b) $\hat{\rho} = \hat{\alpha} + \hat{\beta} = 0.994712$; $\hat{\sigma}^2 = 0.015714/(1 - 0.994712) = 2.972$ (%² per day).
- Daily volatility $\sqrt{2.972} = 1.724\%$; annual $1.724 \times \sqrt{251.4} = 27.3\%$, against the sample value 19.2%.
- Half-life $\ln 0.5 / \ln 0.994712 = -0.6931 / (-0.005302) = 130.7$ days.
- Interpretation: the long-run volatility exceeds the sample value because dividing by $1 - \hat{\rho} = 0.005288$ magnifies small errors in $\hat{\rho}$; the next step measures how much.

A3: Long-Run Volatility, Half-Life and Delta Method: Solution (2/2) [Solved]

- (c) Gradient $\nabla \sigma^2 = \left(\frac{1}{1-p}, \frac{\omega}{(1-p)^2}, \frac{\omega}{(1-p)^2} \right) = (189.1, 562.0, 562.0)$; $\widehat{\text{s.e.}}(\hat{\sigma}^2) = \sqrt{\nabla' \hat{V} \nabla} = 2.35$.
- Annual volatility $v = \sqrt{251.4 \sigma^2}$ has $\partial v / \partial \sigma^2 = 251.4 / (2v)$, so $\widehat{\text{s.e.}}(\hat{v}) = 10.8$ points around 27.3%: a 95% band of about ± 21 points.
- (d) $|\partial h / \partial p| = -\ln 0.5 / [p(\ln p)^2] = 24\,789$, so $\widehat{\text{s.e.}}(\hat{h}) = 24\,789 \times 0.0047 = 116$ days for $\hat{h} = 130.7$ days.
- A symmetric interval $\hat{h} \pm 1.96 \times 116$ would include negative half-lives: the linearisation fails because $h(p)$ explodes as $p \rightarrow 1$.
- Instead, h is increasing in p : map $\hat{p} \pm 1.96 \times 0.0047 = [0.9855; 1.0039]$ to $h \in [48; \infty)$, since the upper end passes $p = 1$.
- Interpretation: the data bound the half-life from below only; whether $p = 1$ is plausible needs a proper test on the boundary (B7).

A4: Finiteness of the Fourth Moment (1/2) [Proposed]

Solution discussed in class

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- GARCH(1,1) with Normal innovations (GARCH-N): $\hat{\alpha} = 0.120604$, $\hat{\beta} = 0.859878$; sample kurtosis of the returns 13.7
- GARCH(1,1)-t: $\hat{\alpha} = 0.1226$, $\hat{\beta} = 0.8721$, $\hat{\nu} = 6.26$; with Normal z_t , $K = 3[1 - (\alpha + \beta)^2]/[1 - (\alpha + \beta)^2 - 2\alpha^2]$ if the denominator is positive (Seminar 1, A9)

What to do

- (a) Check the condition $(\alpha + \beta)^2 + 2\alpha^2 < 1$ for GARCH-N and compute the implied kurtosis K .
- (b) Repeat (a) with $\hat{\alpha}$ and $\hat{\beta}$ rounded to 3 decimals and compare both values of K with the sample kurtosis.
- (c) For Student-t innovations, compute $\kappa_z = 3(\nu - 2)/(\nu - 4)$ and check the condition $(\alpha + \beta)^2 + (\kappa_z - 1)\alpha^2 < 1$ for the GARCH-t estimates.

A4: Finiteness of the Fourth Moment (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report both conditions, the two values of K and κ_z . Interpretation: can the kurtosis of the returns be used to validate a fitted GARCH-t?
- Model: A3

A5: Robust versus Classical Variance of the Gaussian QMLE, ARCH(1) (1/2) [Proposed]

Solution discussed in class

Context and data

- ▣ ARCH(1): $\varepsilon_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2$ with ω known; z_t i.i.d., mean 0, variance 1, kurtosis $\kappa_z = \mathbb{E} z_t^4 < \infty$, not necessarily Normal
- ▣ Gaussian quasi-log-likelihood per day $\ell_t(\alpha) = -\frac{1}{2}(\ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2)$; notation $x_t = \varepsilon_{t-1}^2 / \sigma_t^2$. Data for (d): the S&P 500 GARCH-N residual kurtosis 4.73 (Lecture 5)

What to do

- ▣ (a) Derive the score $s_t = \partial \ell_t / \partial \alpha$ and its variance $B = \mathbb{E} s_t^2$.
- ▣ (b) Derive the second derivative $\partial^2 \ell_t / \partial \alpha^2$ and $A = -\mathbb{E} \partial^2 \ell_t / \partial \alpha^2$.
- ▣ (c) Show that the sandwich variance $A^{-1} B A^{-1}$ equals $\frac{\kappa_z - 1}{2} A^{-1}$, where A^{-1} is the classical (inverse-Hessian) variance.
- ▣ (d) Compute the ratio of robust to classical standard errors for Student-t innovations with $\nu = 6.26$ and for $\hat{\kappa}_z = 4.73$.

A5: Robust versus Classical Variance of the Gaussian QMLE, ARCH(1) (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report s_t , A , B , the variance ratio and the two numerical ratios; compare with the observed ratio of 1.37 in Lecture 5.
- Interpretation: what happens to this ratio, and to the $\sqrt{n}$ theory, when $\kappa_z = \infty$? Model: A3

A6: Autocorrelation of Squared Shocks (1/2) [Proposed]

Solution discussed in class

Context and data

- GARCH(1,1) with $\mathbb{E}\varepsilon_t^4 < \infty$ is an ARMA(1,1) in ε_t^2 : $\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + \nu_t - \beta\nu_{t-1}$, with $\nu_t = \varepsilon_t^2 - \sigma_t^2$ (Lecture 5)
- For an ARMA(1,1) $y_t = \phi y_{t-1} + u_t + \theta u_{t-1}$: $\rho_1 = (1 + \phi\theta)(\phi + \theta)/(1 + 2\phi\theta + \theta^2)$ and $\rho_k = \phi^{k-1}\rho_1$
- Data for (b): S&P 500 GARCH-N, $\hat{\alpha} = 0.1206$, $\hat{\beta} = 0.8599$; sample autocorrelations of the squared demeaned returns 0.313 (lag 1) and 0.248 (lag 10)

What to do

- (a) Substitute $\phi = \alpha + \beta$ and $\theta = -\beta$ and derive $\rho_1(\varepsilon^2) = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$ and $\rho_k = \rho_1(\alpha + \beta)^{k-1}$.
- (b) Evaluate ρ_1 and ρ_{10} for the S&P 500 estimates and compare them with the sample values.
- (c) Explain why the gap can be large even if the model is correct, using the moment condition of A4.

A6: Autocorrelation of Squared Shocks (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report the formula, ρ_1 , ρ_{10} and the sample values. Interpretation: is matching the autocorrelation function of r_t^2 a strong or a weak check of a GARCH fit?
- Model: A1

B1: GARCH-t on the S&P 500 with Diagnostics (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- Mean equation $r_t = \mu + \varepsilon_t$; variance equation GARCH(1,1) [Bollerslev \(1986\)](#); innovations Student-t with ν degrees of freedom, scaled to unit variance

What to do

- (a) Test for ARCH effects before modelling: Ljung–Box $Q(10)$ on r_t^2 and Engle's LM test with 5 lags on the demeaned returns, at the 5% level.
- (b) Estimate GARCH(1,1)-t by maximum likelihood (`arch_model(r, p=1, q=1, dist='t')`); report robust [Bollerslev & Wooldridge \(1992\)](#) and classical standard errors and their ratio.
- (c) Diagnose $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ at 5%: $Q(10)$ on $\hat{z}_t$ and $\hat{z}_t^2$, LM(5), and the three sign-bias t -tests plus their joint χ^2_3 test [Engle & Ng \(1993\)](#).

B1: GARCH-t on the S&P 500 with Diagnostics (2/2) [Solved]

What to report and interpret

- Report a table of estimates with both standard errors and a table of test statistics with p -values and decisions.
- Interpretation: which diagnostics pass and which fail, what does the failure suggest, and why are the robust standard errors larger?

B1: GARCH-t on the S&P 500 with Diagnostics: Solution (1/2) [Solved]

	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\nu}$	$\hat{\alpha} + \hat{\beta}$
Estimate	0.0157	0.1226	0.8721	6.26	0.9947
Robust s.e.		0.0107	0.0103	0.48	
Classical s.e.		0.0101	0.0096		

- (a) Before modelling, $Q(10) = 6133$ and $LM(5) = 1684$ against 5% critical values 18.3 (χ^2_{10}) and 11.1 (χ^2_5); both $p < 0.001$, so no-ARCH is rejected.
- (b) Reading the table: $\hat{\beta} = 0.8721$ carries yesterday's variance, $\hat{\alpha} = 0.1226$ adds the news, $\hat{\nu} = 6.26$ signals heavy tails; persistence 0.9947.
- The robust/classical ratio is 1.06 for $\hat{\alpha}$ and 1.08 for $\hat{\beta}$: the Student-t likelihood is itself misspecified (skewness -0.58), so the sandwich is larger.

B1: GARCH-t on the S&P 500 with Diagnostics: Solution (2/2) [Solved]

- (c) Variance equation: $Q(10)$ on $\hat{z}_t^2 = 13.6$ ($p = 0.192$) and $LM(5) = 5.8$ ($p = 0.326$): not rejected at 5%: no remaining clustering is detected (naive χ^2 laws, see the caveat below).
- Mean equation: $Q(10)$ on $\hat{z}_t = 21.9$ ($p = 0.016$): rejected at 5%; a small autocorrelation remains that a constant mean ignores.
- Asymmetry: sign bias $t = 3.56$, negative size bias $t = 1.13$, positive size bias $t = -2.61$; joint $\chi^2_3 = 40.7$ against 7.81: rejected, asymmetry is left in the residuals.
- Caveat: with estimated parameters these χ^2 laws are only approximate [Li & Mak \(1994\)](#); here the rejections are far from the critical values, so the decisions stand.
- Interpretation: the variance dynamics are captured, but a symmetric GARCH misses the leverage effect; the next step is GJR-GARCH (B2).

B2: GJR vs GARCH on the S&P 500 (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- GJR-GARCH(1,1)-t [Glosten, Jagannathan & Runkle \(1993\)](#): $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, $I_{t-1} = 1(\varepsilon_{t-1} < 0)$; GARCH(1,1)-t of B1 is the special case $\gamma = 0$

What to do

- (a) Estimate GJR-GARCH(1,1)-t and report $\hat{\alpha}$, $\hat{\gamma}$ with its robust t -statistic, and $\hat{\beta}$.
- (b) Test $H_0: \gamma = 0$ with $LR = 2(\ell_{GJR} - \ell_{GARCH})$ against χ_1^2 at 5%.
- (c) Compare AIC and BIC of the two models and repeat the joint sign-bias test of B1 on the GJR residuals.
- (d) Measure the economic size of the asymmetry with the news impact curve: next-day variance after shocks of -3% and $+3\%$.

What to report and interpret

- Report the estimates, the LR test (statistic, p -value, decision), ΔAIC , ΔBIC and the two news-impact values.

B2: GJR vs GARCH on the S&P 500 (2/2) [Solved]

What to report and interpret

- Interpretation: why is $\hat{\alpha}$ exactly 0, and is the χ^2_1 law of the LR test still valid?

B2: GJR vs GARCH on the S&P 500: Solution [Solved]

- (a) $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ (robust $t = 10.03$), $\hat{\beta} = 0.881$: good news has slope $\hat{\alpha}$, bad news $\hat{\alpha} + \hat{\gamma}$.
- (b) $LR = 2(-8944.6 - (-9062.5)) = 235.65$, $p < 0.001$, far above 3.84: H_0 is rejected at 5%; the χ^2_1 law assumes a correct Student-t density (B1 finds skewness), but the robust t of (a) gives the same verdict.
- (c) $\Delta AIC = -233.7$ and $\Delta BIC = -226.8$ (GJR minus GARCH): both prefer GJR; the joint sign-bias χ^2_3 falls from 40.7 to 28.1, still above 7.81.
- (d) From the sample variance 1.47, a -3% shock gives next-day variance 3.16 and a $+3\%$ shock 1.31: the forecast after bad news is 2.4 times that after good news, and the $+3\%$ shock even lowers the variance ($\hat{\alpha} = 0$).
- $\hat{\alpha} = 0$ sits on the bound $\alpha \geq 0$: positive shocks do not raise volatility; $\gamma = 0$ is interior, so χ^2_1 holds if the true $\alpha > 0$, and if $\alpha_0 = 0$ the law is non-standard [Andrews \(2001\)](#), yet $LR = 235.65$ rejects under any reasonable law.

B3: GJR vs GARCH on the BET [Proposed]

Solution discussed in class

Context and data

- BET (data/market/BET.csv), daily closes 2000–2026, 6 670 daily log returns in %
- The same two models as in B2: GARCH(1,1)-t and GJR-GARCH(1,1)-t, constant mean, Student-t innovations

What to do

- (a) Estimate both models and report $\hat{\gamma}$ with its robust t -statistic.
- (b) Test $H_0: \gamma = 0$ by LR against χ^2_1 at the 5% and 1% levels.
- (c) Compute ΔAIC and ΔBIC (GJR minus GARCH) and the BIC penalty per parameter, $\ln n$.

What to report and interpret

- Report the estimates, the LR decision at both levels and the two criteria. Interpretation: do LR and BIC agree, why can they disagree, and is the leverage effect weaker on the BET than on the S&P 500?
- Model: B2

B4: Innovation Distribution for Bitcoin (1/2) [Proposed]

Solution discussed in class

Context and data

- Bitcoin (data/market/BTC-USD.CC.csv), daily closes 2014–2026, 4 384 daily log returns in %, 365 days a year
- GARCH(1,1) with constant mean; four innovation laws: Normal, Student-t, skewed-t [Hansen \(1994\)](#)(asymmetry λ) and GED

What to do

- (a) Estimate the four models and compare their log-likelihoods and BIC.
- (b) Test $H_0: \lambda = 0$ (skewed-t against Student-t) by LR against χ^2_1 at 5%.
- (c) Report the applicable shape and skewness parameters with their standard errors, and $\hat{\alpha} + \hat{\beta}$, for each law; the Normal distribution has none, and the GED shape is not a degrees-of-freedom parameter; state which parameters sit at a bound.
- (d) Test the Normal distribution ($1/\nu = 0$) against the Student-t distribution by LR with the boundary law $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (5% critical value 2.71).

B4: Innovation Distribution for Bitcoin (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report a table of ℓ , BIC and persistence, and the two LR tests. Interpretation: the sample skewness of $\hat{z}_t$ is negative, so why does the likelihood find no skewness? With $\hat{\nu} < 4$, is Gaussian-QMLE inference valid for Bitcoin [Hall & Yao \(2003\)](#), and how does it differ from inference for the Student-t MLE of B1?
- Model: B1, B2

B5: Forecast Comparison on the S&P 500 (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$; forecast period 2016–2026 (2 692 one-day-ahead forecasts)
- Models: GARCH-N, GARCH-t, GJR-t, EGARCH-t re-estimated every January on all data since 2000 (expanding window); EWMA with $\lambda = 0.94$ J.P. Morgan/Reuters (1996); Hist-20, the mean of the last 20 squared returns
- Proxy of the true variance: r_t^2 , ignoring the mean ($\hat{\mu}^2 = 0.0062$ against an average variance of 1.47); loss QLIKE $= r_t^2 / h_t + \ln h_t$ Patton (2011), with h_t the variance forecast

What to do

- (a) Compute the mean QLIKE of each model.
- (b) Test equal accuracy against GARCH-t with the Diebold–Mariano t -statistic on $d_t = L_t^m - L_t^{GARCH-t}$, HAC standard error with 5 lags, at 5% Diebold & Mariano (1995).
- (c) Run the Mincer–Zarnowitz regression $r_t^2 = a + b h_t + u_t$ Mincer & Zarnowitz (1969) with HAC errors and the Wald test of $a = 0$, $b = 1$.

B5: Forecast Comparison on the S&P 500 (2/2) [Solved]

What to do

- (d) Repeat the comparison with a rolling 1 000-day window, re-estimated each January, and the Giacomini–White test [Giacomini & White \(2006\)](#) (instruments 1, d_t).
- (e) Compute the 90% Model Confidence Set [Hansen, Lunde & Nason \(2011\)](#) on the QLIKE losses of the expanding-window forecasts of (a): statistic T_R , stationary bootstrap with mean block length 10, 10 000 draws, fixed seed.

What to report and interpret

- Report two tables: QLIKE, DM t , MZ b , R^2 and Wald p ; then GW statistic, p -value and MCS p -value.
- Interpretation: which models are significantly worse than GARCH-t, why is R^2 low, why is DM not valid for GJR-t vs GARCH-t, and do DM and MCS agree?

B5: Forecast Comparison on the S&P 500: Solution (1/2) [Solved]

Model	QLIKE	DM t	MZ b	MZ R^2	Wald p
GARCH-N	0.7510	-1.47	0.97	0.250	0.995
GARCH-t	0.7570	—	0.88	0.238	0.804
GJR-t	0.7319	-1.22	0.84	0.276	0.680
EGARCH-t	0.7454	-0.36	1.24	0.293	0.713
EWMA	0.8149	3.48	0.89	0.172	0.685
Hist-20	0.8888	4.10	0.68	0.152	0.028

- Procedure: forecast h_t each day with parameters from the last January, compute L_t , average over 2 692 days, and regress d_t on a constant with HAC errors.
- (a–b) Lowest QLIKE: GJR-t; only EWMA ($t = 3.48$) and Hist-20 ($t = 4.10$) exceed 1.96: they are significantly worse at 5%.
- (c) Unbiasedness is not rejected for the GARCH models (Wald $p > 0.6$) but is rejected for Hist-20 ($p = 0.028$); R^2 is low because $r_t^2 \approx \sigma_t^2 z_t^2$ is a noisy proxy.

B5: Forecast Comparison on the S&P 500: Solution (2/2) [Solved]

Model	t (rolling)	GW	p	MCS p
GARCH-N	1.87	5.85	0.054	0.586
GARCH-t	—	—	—	0.353
GJR-t	−0.18	2.49	0.288	1.000
EGARCH-t	0.07	3.21	0.201	0.586
EWMA	3.20	11.40	0.003	0.008
Hist-20	3.87	16.41	< 0.001	0.001

- (d) Why GW: with nested models, estimated parameters and an expanding window the DM differential degenerates under H_0 , so DM is not $N(0, 1)$; a rolling window keeps estimation noise and makes GW valid.
- $GW = n\bar{Z}'\hat{\Omega}^{-1}\bar{Z} \sim \chi_2^2$ (critical value 5.99): rejected for EWMA and Hist-20 only; GARCH-N is borderline ($p = 0.054$).
- (e) MCS on the expanding-window forecasts: a model stays in the 90% set if its MCS p -value exceeds 0.10; the set is GARCH-N, GARCH-t, GJR-t, EGARCH-t, the same verdict as DM, now with the error rate controlled over six models.

B6: Forecast Comparison on the BET (1/2) [Proposed]

Solution discussed in class

Context and data

- BET (data/market/BET.csv), daily closes 2000–2026, 6 670 daily log returns in %; forecast period 2016–2026 (2 675 one-day-ahead forecasts)
- The same six models, re-estimation scheme, proxy r_t^2 and QLIKE loss as in B5

What to do

- (a) Compute the mean QLIKE of each model and the Diebold–Mariano t -statistic against GARCH-t (HAC, 5 lags), at 5%.
- (b) Run the Mincer–Zarnowitz regression and the Wald test of $a = 0$, $b = 1$ for each model.
- (c) Compute the Giacomini–White test with a rolling 1 000-day window and, as in B5 (e), the 90% Model Confidence Set on the expanding-window QLIKE losses.

What to report and interpret

- Report the two tables of B5 for the BET. Interpretation: does the ranking change between the S&P 500 and the BET, do DM and MCS give the same verdict for EWMA, and which part of the BET model would you improve first?

B6: Forecast Comparison on the BET (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Model: B5

B7: How Precise Is the Half-Life? Bootstrap and IGARCH Test (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- GARCH(1,1)-t of B1: $\hat{p} = \hat{\alpha} + \hat{\beta} = 0.9947$, half-life 131 days; estimation imposes $\alpha + \beta \leq 1$; random seed fixed for reproducibility

What to do

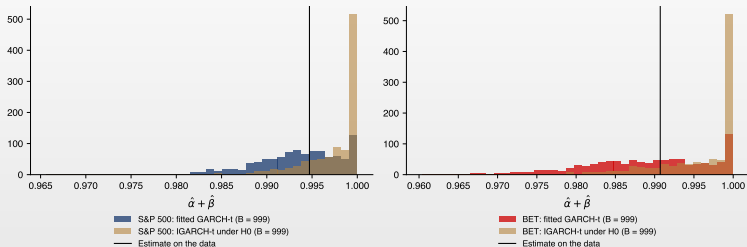
- (a) Simulate $B = 999$ paths of 6 714 days (500 burn-in days) from the fitted GARCH(1,1)-t and re-estimate the model on each path.
- (b) Compute the 95% percentile intervals of $\hat{p}^*$ and of the half-life, the bootstrap standard deviation of $\hat{p}^*$ and the share of draws at the bound $p = 1$; compare with the robust standard error.
- (c) Test $H_0: \alpha + \beta = 1$: fit IGARCH(1,1)-t, compute $LR = 2(\ell_{GARCH} - \ell_{IGARCH})$, simulate $B = 999$ paths **under** H_0 , re-estimate both models on each, and compute the bootstrap p -value $(1 + \#\{LR^* \geq LR\})/(B + 1)$.
- (d) Compare this p -value with the naive χ_1^2 and the boundary mixture $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$; decide at 5% and 10%.

B7: How Precise Is the Half-Life? Bootstrap and IGARCH Test (2/2) [Solved]

What to report and interpret

- Report the histogram of $\hat{p}^*$ under both schemes, the intervals, the three p -values and the decision.
- Interpretation: why is the half-life interval so asymmetric, and why is “the percentile interval reaches 1” not a test of IGARCH?

B7: How Precise Is the Half-Life? Bootstrap and IGARCH Test: Solution (1/2) [Solved]



- (a–b) Blue: $\hat{\rho}^*$ from the fitted model; 95% interval [0.9833; 1.0000]; bootstrap s.d. 0.0047 against the robust s.e. 0.0047: close.
- Half-life from 41 days to ∞ : $\ln 0.5 / \ln x$ explodes as $x \rightarrow 1$ (delta-method s.e. 116 days, A3); 8.9% of draws sit at the bound, so the interval is truncated at 1 [Andrews \(2000\)](#).

B7: How Precise Is the Half-Life? Bootstrap and IGARCH Test: Solution (2/2) [Solved]

- (c) IGARCH-t fit: $\hat{\alpha}_0 = 0.127$, $\hat{\nu}_0 = 6.03$; $LR = 1.11$.
- Under H_0 (amber histogram), 52% of the re-estimates sit at $p = 1$, close to the half predicted by the mixture law; the simulated 95% critical value of LR is 3.36.
- (d) p -values: 0.29 (naive χ^2_1), 0.15 (mixture), 0.17 (bootstrap LR) and 0.17 (bootstrap $\hat{p}$).
- Decision: H_0 is not rejected at 5% or at 10%; the naive χ^2_1 roughly doubles the p -value, the mixture and the bootstrap agree.
- Economic conclusion: 27 years of data cannot tell a half-life of about 130 days from an infinite one; simulating from $\hat{\theta}$ describes precision, simulating from the null model is what a test needs.

B8: Bootstrap and IGARCH Test for the BET [Proposed]

Solution discussed in class

Context and data

- BET (data/market/BET.csv), daily closes 2000–2026, 6 670 daily log returns in %
- GARCH(1,1)-t with $\hat{p} = 0.9907$ (half-life 74 days); the same procedure and number of draws as in B7

What to do

- (a) Run the parametric bootstrap with $B = 999$ from the fitted model; report the 95% intervals for p and for the half-life and the share of draws at $p = 1$.
- (b) Compare the bootstrap standard deviation of $\hat{p}^*$ with the robust standard error of $\hat{\alpha} + \hat{\beta}$.
- (c) Test IGARCH by simulation under H_0 ($B = 999$) and compare with the mixture p -value, at 5%.

What to report and interpret

- Report the intervals, the two standard errors and the test. Interpretation: is the BET half-life shorter than the S&P 500 one with confidence, and what would a formal test of the difference need?
- Model: B7

C1: Is the Romanian Market's Volatility Driven by the US? [Proposed]

Solution discussed in class

The question

- Wall Street closes after the Bucharest Stock Exchange: yesterday's S&P 500 shock is news for today's BET
- Does u_{t-1}^2 , the squared lagged S&P 500 return, raise the conditional variance of the BET?
- Model: $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta u_{t-1}^2$, with $r_t = \mu + \phi u_{t-1} + \varepsilon_t$

Deliverable

- A Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should contain

- Prices joined on common trading days first, then returns
- The likelihood written step by step (Student-t), estimated by numerical optimisation
- An LR test of $\delta = 0$; note that $\delta \geq 0$ puts the null on the boundary
- Stability: two subperiods (e.g. before and after 2013)
- The model-based share of the BET variance generated by the US term, $\sum_{j \geq 0} \beta^j \delta u_{t-1-j}^2 / \sigma_t^2$, and how it changes over time

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “px holds daily S&P 500 closes. Fit a GARCH(1,1) with Student- t innovations using arch and give me the annualised long-run volatility and the half-life of a shock.”

The assistant's answer

```
import numpy as np
from arch import arch_model
r = np.log(px).diff().dropna()          # daily log returns
am = arch_model(r, vol='GARCH', p=1, q=1, dist='t')
res = am.fit(displ='off')
om = res.params['omega']
a, b = res.params['alpha[1]'], res.params['beta[1]']
lr_var = om / (1 - a - b)               # long-run daily variance
ann_vol = lr_var * 252                  # annualised volatility
half_life = np.log(0.5) / np.log(b)    # half-life in days
```

- “For the S&P 500 this gives $\text{ann_vol} = 0.075$ (7.5% a year) and a half-life of 5.1 days.”

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- The code runs, but the answer contains **three** mistakes typical for GARCH work
 - ▶ Everything else in the answer is correct
- For each mistake
 - ▶ Name the line and say why it is wrong
 - ▶ Say how you would detect it: a warning, a number from A3, a slide of the lecture
 - ▶ Write the corrected line(s) and the corrected numbers
- Log each mistake as in `AI_ERRORS.md`
 - ▶ What was asked, what the AI answered, what was wrong, how you found out, the correct version
- Hint: compare with A3 and with the slide “Common Pitfalls”

C3: Intraday Entropy and Bitcoin's Value at Risk (1/2) [Proposed]

Solution discussed in class

Context

- Lecture 5 case study: [Pele & Mazurencu-Marinescu-Pele \(2019\)](#) forecast Bitcoin's Value at Risk (VaR), $\text{VaR}_\alpha = -q_\alpha(r_t)$, from the entropy S_t of intraday returns and backtest it against GARCH models
- Question: does S_{t-1} still add information beyond a GARCH-t forecast after 2018?

What to report

- A table by model, VaR level and period: violation rate, LR_{uc} , LR_i , LR_{full} and their p -values
- The mean quantile loss $(\alpha - I_t)(r_t + \widehat{\text{VaR}}_t)$ of each model and a Diebold–Mariano test against GARCH-t
- $\hat{\gamma}$ with a robust standard error and the LR test, taking into account that $\gamma \geq 0$ puts the null on the boundary
- One slide: does entropy add information beyond GARCH-t, and the main limitation of your evidence
- Model: B1 (GARCH-t), B2 (LR test), B5 (out-of-sample comparison)

C3: Intraday Entropy and Bitcoin's Value at Risk (2/2) [Proposed]

Solution discussed in class

Tasks

1. Load the Bitcoin daily closes from `data/market/BTC-USD.CC.csv`, compute daily log returns and split them into the paper's sample (8 October 2015 – 2 October 2018) and the period after it, up to 18 September 2026.
2. On a rolling window of 250 days, forecast next-day VaR 1% and VaR 5% with historical simulation, GARCH(1,1)-N, GARCH(1,1)-t and GJR-GARCH(1,1)-t, re-estimating every day.
3. For each model, level and period, compute the violation rate and the Christoffersen (1998) statistics LR_{uc} , LR_i , LR_{full} with p -values, and compare 2015–2018 with Tables 4–5 of the paper.
4. From the 5-minute bars in `data/market/intraday/BTC-USD.CC_5m.csv` (September 2024 – September 2026), compute S_t for each UTC day exactly as in Eqs. 2–4 of the paper, which uses one-minute prices.
5. On the same windows, forecast Entropy VaR and Entropy AR VaR (Eqs. 11–14) and fit a GARCH-t with γS_{t-1} added to the variance equation; test $\gamma = 0$.

Common Pitfalls

- ▣ Fitting GARCH to returns in decimals: the optimiser may stop far from the maximum; use % or rescale
- ▣ Reading classical standard errors when the innovations are not Normal; use robust ones
- ▣ Trusting long-run volatility and half-life when $\alpha + \beta \approx 1$ without an interval
- ▣ Evaluating forecasts in sample, or with losses that are not robust to a noisy proxy
- ▣ Using an LR test for a parameter on the boundary (e.g. $\delta = 0$ with $\delta \geq 0$) with the usual χ^2 critical value, or reading a truncated bootstrap interval as a test
- ▣ Joining two markets on returns instead of prices: creates spurious multi-day returns

Take-Home Assignment (Team Project)

- ▣ Choose one series not covered in class (another index, an exchange rate, a commodity or a crypto-asset)
- ▣ Test for ARCH effects; fit GARCH, GJR and EGARCH with three innovation distributions; select by BIC and diagnostics
- ▣ Evaluate one-day and one-month forecasts out of sample against EWMA with QLIKE and Diebold–Mariano
- ▣ Report the half-life with a bootstrap interval; for an entropy-based alternative, see [Pele, Lazar & Dufour \(2017\)](#)
- ▣ Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Statistics of Financial Markets
- ☐  Applied Time Series Analysis with Python

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_05

References (1/3)

- ▣ Andrews, D.W.K. (2000). Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space. *Econometrica*, 68(2), 399–405.
- ▣ Andrews, D.W.K. (2001). Testing when a parameter is on the boundary of the maintained hypothesis. *Econometrica*, 69(3), 683–734.
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- ▣ Bollerslev, T., Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.
- ▣ Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R.F., Ng, V.K. (1993). Measuring and testing the impact of news on volatility. *Journal of Finance*, 48(5), 1749–1778.

References (2/3)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., White, H. (2006). *Tests of conditional predictive ability*. *Econometrica*, 74(6), 1545–1578.
- ▣ Glosten, L.R., Jagannathan, R., Runkle, D.E. (1993). *On the relation between the expected value and the volatility of the nominal excess return on stocks*. *Journal of Finance*, 48(5), 1779–1801.
- ▣ Hall, P., Yao, Q. (2003). *Inference in ARCH and GARCH models with heavy-tailed errors*. *Econometrica*, 71(1), 285–317.
- ▣ Hansen, B.E. (1994). *Autoregressive conditional density estimation*. *International Economic Review*, 35(3), 705–730.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). *The model confidence set*. *Econometrica*, 79(2), 453–497.
- ▣ J.P. Morgan/Reuters (1996). *RiskMetrics — Technical Document*, 4th ed. New York.

References (3/3)

- ▣ Li, W.K., Mak, T.K. (1994). On the squared residual autocorrelations in non-linear time series with conditional heteroskedasticity. *Journal of Time Series Analysis*, 15(6), 627–636.
- ▣ Mincer, J., Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (ed.), *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, NBER.
- ▣ Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily Value at Risk. *Entropy*, 21(2), 102.